

Project Overview

An end to end data analysis project examining the performance of The Daily Grind - D2C (Coffee & Coffeeware) business to understand how rising COGS has impacted profit margin over the years (2023-2025) and to identify the products that require prompt attention to price modification.

Source

This analysis uses the 'Data Analysis Project November 2025 Youtube.zip' dataset, sourced from Github user Gaelim (accessed in August 2026).

The dataset is synthetic, built to resemble a real-world e-commerce business, and features a fictitious brand, **The Daily Grind**, a coffee-focused D2C (direct-to-consumer) brand selling coffee products alongside brewing equipment, accessories, merchandise, and subscription offerings. The temporal scope of this dataset ranges from 2nd January 2023 until 30th November 2025 and doesn't contain December 2025 data.

Note: This dataset does not reflect any real company's actual sales or pricing catalog.

Tools used

1. SQL Server

- Data cleaning (to exclude NULLs and deduplicate IDs)
- Data preparation (Deriving new columns and replacing NULL rows with calculated values)
- EDA to get familiar with the dataset and business context..
- Building business ready objects such as Facts and Dimension through Views and Tables which then could be imported and used for visualization.
- To store all the related source tables in "The Daily Grind" Database and to access it efficiently.

2.PowerBI - DAX for KPI's and Metrics, Data Modeling and Visualization

3.Excel - Check datatypes and data consistency of each columns before importing into SQL

Data preparation

1. Trim checks for Text and Alphanumeric columns - no issues found
2. Checked for outliers in dates - no issues found
3. Checked for outliers, negatives and zeros in numeric columns. - no issues found
4. Upon using various checks to detect data quality issues, I found that only CustomerID and Revenue in Orders table had NULL values.
5. Customer ID with NULL values were excluded considering the immateriality of it to the total Revenue and Total Row Count while Revenue with NULL values were replaced by product of Price and Quantity.

Also found that the OrderDate column in the Orders 2023 and Orders 2024 tables had a consistent date format ("dd-mm-yyyy") and were imported as dates into SQL. The OrderDate column in the Orders 2025 table, however, contained a mix of dates and text values posing as dates, in various formats such as "mm/dd/yyyy", "m/d/yyyy", and "mm-dd-yyyy". As a result, OrderDate in Orders 2025 was imported as VARCHAR into SQL. This data quality check was performed using Excel's ISNUMBER function.

Once imported as VARCHAR, the OrderDate column was implicitly converted to a proper date format when combined with the 2023 and 2024 tables using a UNION ALL in the all_orders view — SQL Server coerced the VARCHAR column to match the DATE type of the other two tables. Then, a sample of converted 2025 dates was spot-checked against the original values to confirm they were parsed correctly.

6. Since Time Intelligence functions in PowerBI require continuous dates to produce correct results and to visualize the days with no sales or no orders, Dimension Date Table (dim_date) was created using Recursive query and DateTime Functions in SQL Server.

Connection Mode & Data Modeling

All the transformed tables

1. fact_order_table (View)
2. dim_product (View)
3. dim_customers (Table)
4. dim_date (Table)

were imported into PowerBI using **import mode** considering the static data, only a few thousand rows and for faster performance.

After checking for anomalies and errors, Star schema was built with a Single Direction filter flowing from Lookup Tables (Dimension Tables) to Fact Table.

All the columns that represent financial amount were imported as DECIMAL(12,2) into SQL and were assigned fixed-point Decimal datatype in PowerBI

Business Context

1. Since the dataset did not contain any information about discounts or expenses , SUM(Revenue) was referred to as **Total Revenue** rather than Gross or Net Revenue.
2. The term **Gross profit** was used as it directly comes from subtracting SUM(COGS) from SUM(Revenue) .
3. Since **Profit Margin** is a non-additive measure, it was calculated directly from base values , $(\text{Total Revenue} - \text{Total COGS}) / \text{Total Revenue}$ and not calculated as

AVERAGE of profit margin of each row. The same method has been applied to **Markup %** as well.

ProductCategory	Markup %	Profit Margin	Average of Markup%_Exist_col	Average of Profit Margin %_Exist_col
☐ Consumables	165.07%	62.27%	196.21%	57.36%
Cold Brew Concentrate	145.00%	59.18%	145.19%	59.16%
Decaf Blend (1kg)	229.88%	69.69%	230.53%	69.70%
Espresso Bean Sampler Pack	67.94%	40.46%	67.81%	40.33%
Seasonal Blend Light Roast	528.29%	84.08%	527.94%	84.06%
Single-Origin Ethiopian (250g)	60.96%	37.87%	61.25%	37.89%
Total	165.07%	62.27%	196.21%	57.36%

- Unit Cost and Unit Price from the fact_order_table were not loaded to the front view since calculating average of those columns undermined the impact of products that had more orders. Hence, they were calculated as $\text{SUM}(\text{COGS}) / \text{SUM}(\text{Quantity Sold})$ and $\text{SUM}(\text{Revenue}) / \text{SUM}(\text{Quantity Sold})$ respectively.
- Upon checking, Unit Cost has increased every year while Unit Price has always been static.
- CAGR (Compounded Annual Growth Rate) has been calculated to understand the annual growth rate of COGS along with Cumulative growth rate.

Hierarchy Used

Product Hierarchy - Product Category -----> Product Name

Continuous Date Hierarchy - Start of Year ----> Start of Month ----> Start of Week

Miscellaneous

Current File Locale is set to English (US) and autodetect Relationship has been Turned off.